

Machine Learning

Lecture 11: Overfitting, Regularization, Validation (II)

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本节内容

- Validation
- Model selection
- Cross-validation

Validation Set

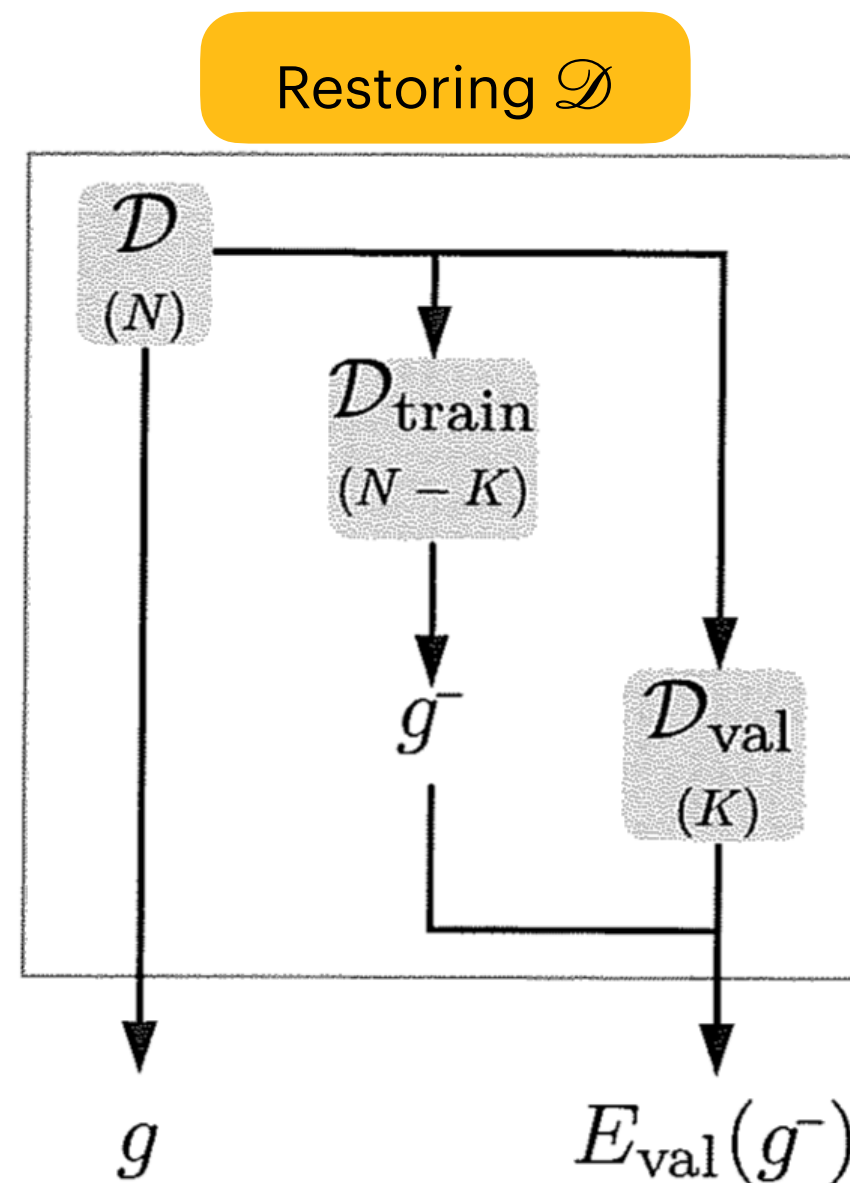
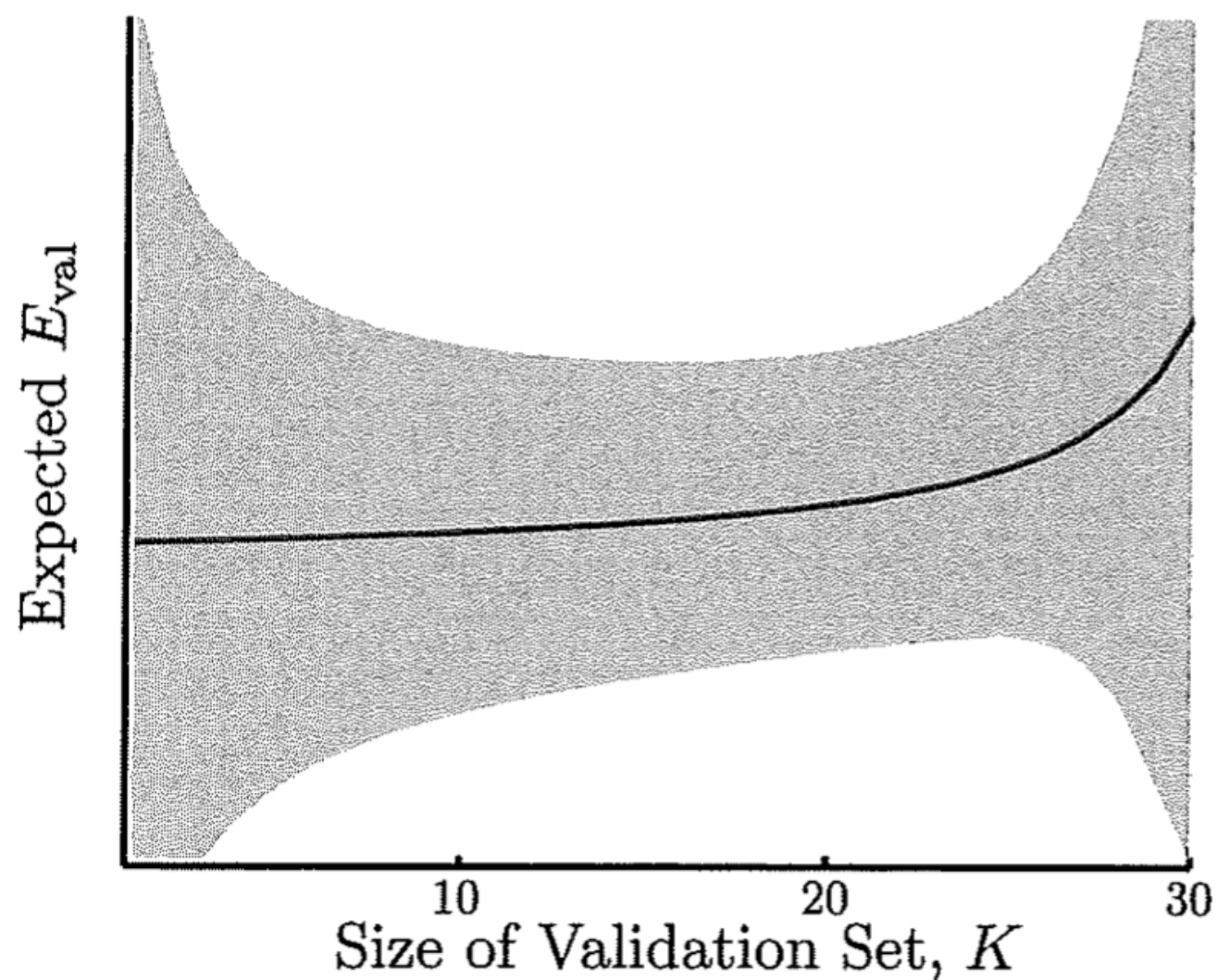
- Validation set is **almost identical but not equivalent** to a test set
- A subset from the data, used to estimate the out-of-sample error
- Validation set is used in training; test set is not used in the training
- $\mathcal{D} = \mathcal{D}_{\text{train}}$ of size $(N - K) \cup \mathcal{D}_{\text{val}}$ of size K

$$\begin{aligned}\mathbb{E}_{\mathcal{D}_{\text{val}}} [E_{\text{val}}(g^-)] &= \frac{1}{K} \sum_{\mathbf{x}_n \in \mathcal{D}_{\text{val}}} \mathbb{E}_{\mathcal{D}_{\text{val}}} [\mathbf{e}(g^-(\mathbf{x}_n), y_n)] \\ &= \frac{1}{K} \sum_{\mathbf{x}_n \in \mathcal{D}_{\text{val}}} \mathbb{E}_{\mathbf{x}_n} [\mathbf{e}(g^-(\mathbf{x}_n), y_n)] \\ &= \frac{1}{K} \sum_{\mathbf{x}_n \in \mathcal{D}_{\text{val}}} E_{\text{out}}(g^-), \\ &= E_{\text{out}}(g^-).\end{aligned}$$

$$E_{\text{out}}(g^-) \leq E_{\text{val}}(g^-) + O\left(\frac{1}{\sqrt{K}}\right)$$

Validation Set

$$E_{\text{out}}(g^-) \leq E_{\text{val}}(g^-) + O\left(\frac{1}{\sqrt{K}}\right)$$



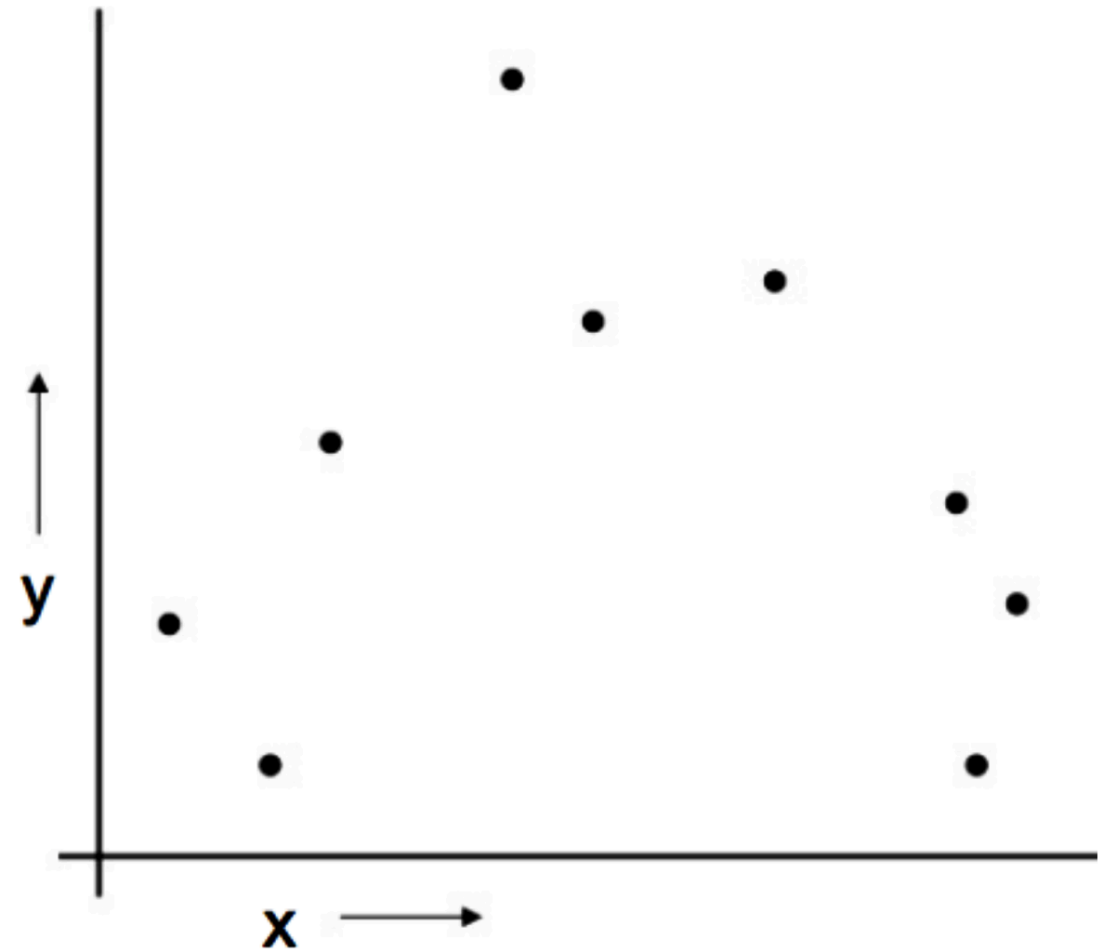
$$E_{\text{out}}(g) \leq E_{\text{out}}(g^-) \leq E_{\text{val}}(g^-) + O\left(\frac{1}{\sqrt{K}}\right)$$

Example:

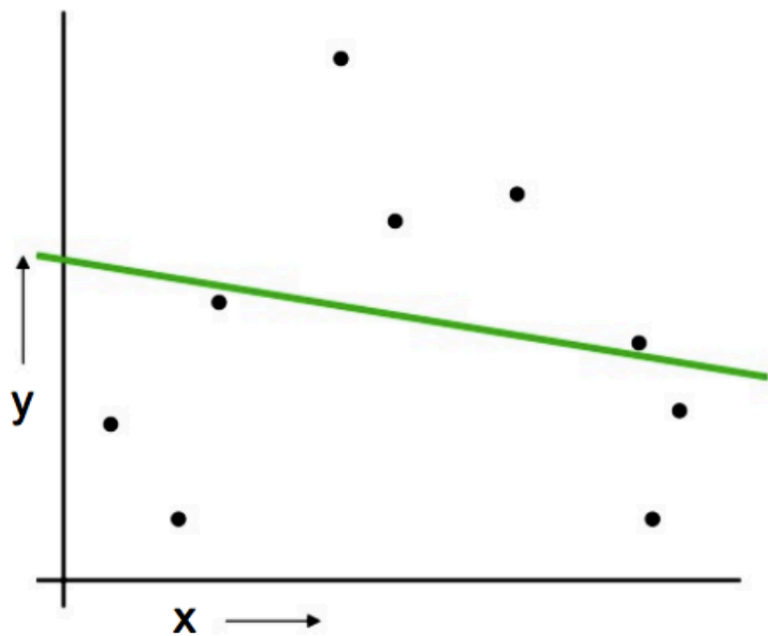
- ▶ How to detect overfitting of your model?
- ▶ What can be observed if your model is overfitting?

$$y = f(x) + \textit{noise}$$

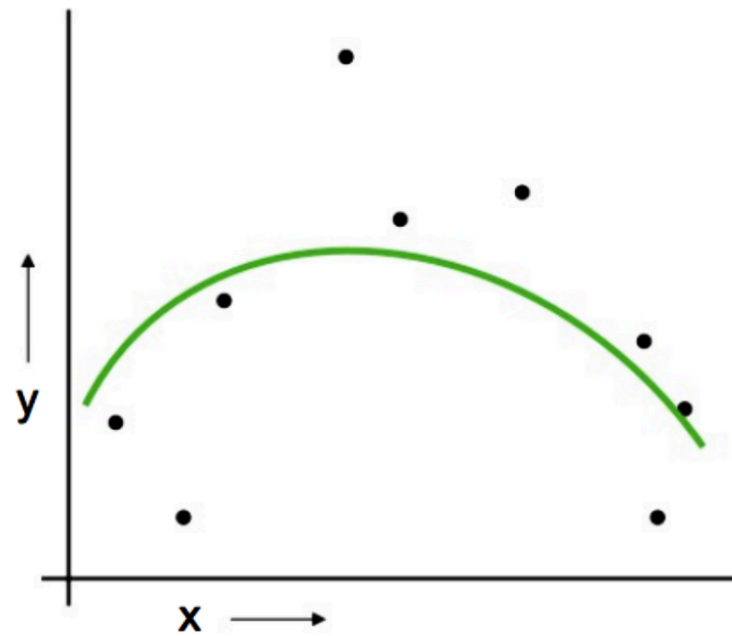
- ▶ Can we learn f from this data?
- ▶ Let's consider 3 methods



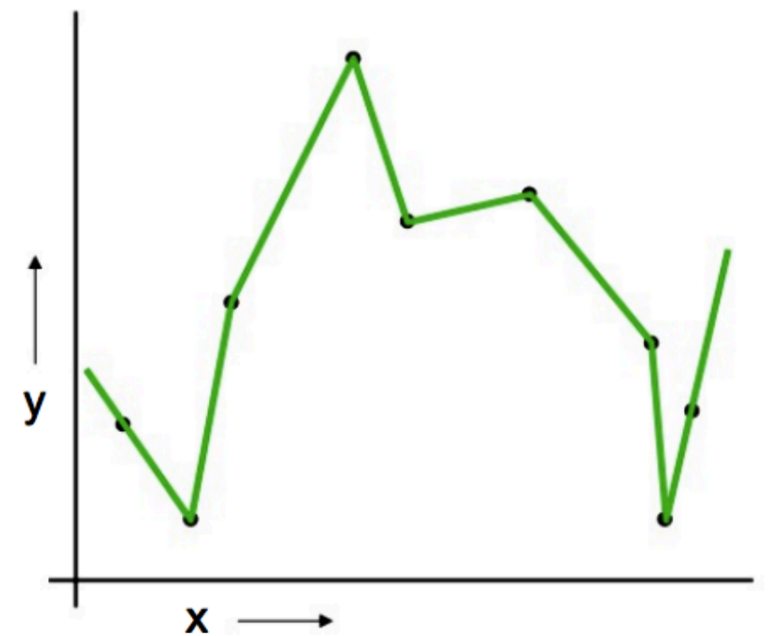
Which model to choose?



Linear Reg.



Quadratic Reg.

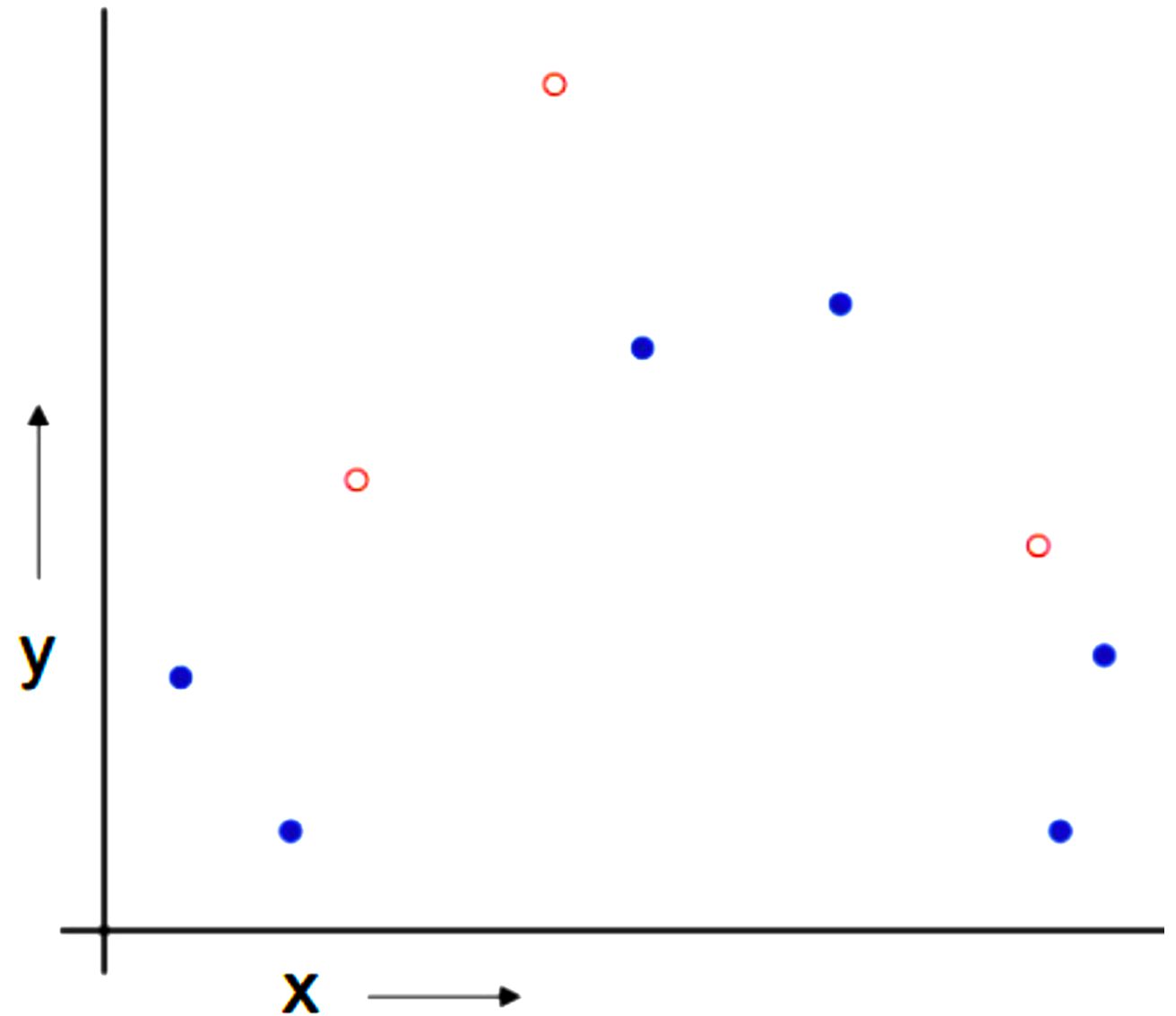


Join-the-dots

- ▶ Which is the best?
- ▶ Why not choose the method with the best fit to the data?
- ▶ How well are you going to predict future data drawn from the same distribution?

The test set method

Randomly choose 30% of the data to be in a **test set**
The remainder is a **training set**

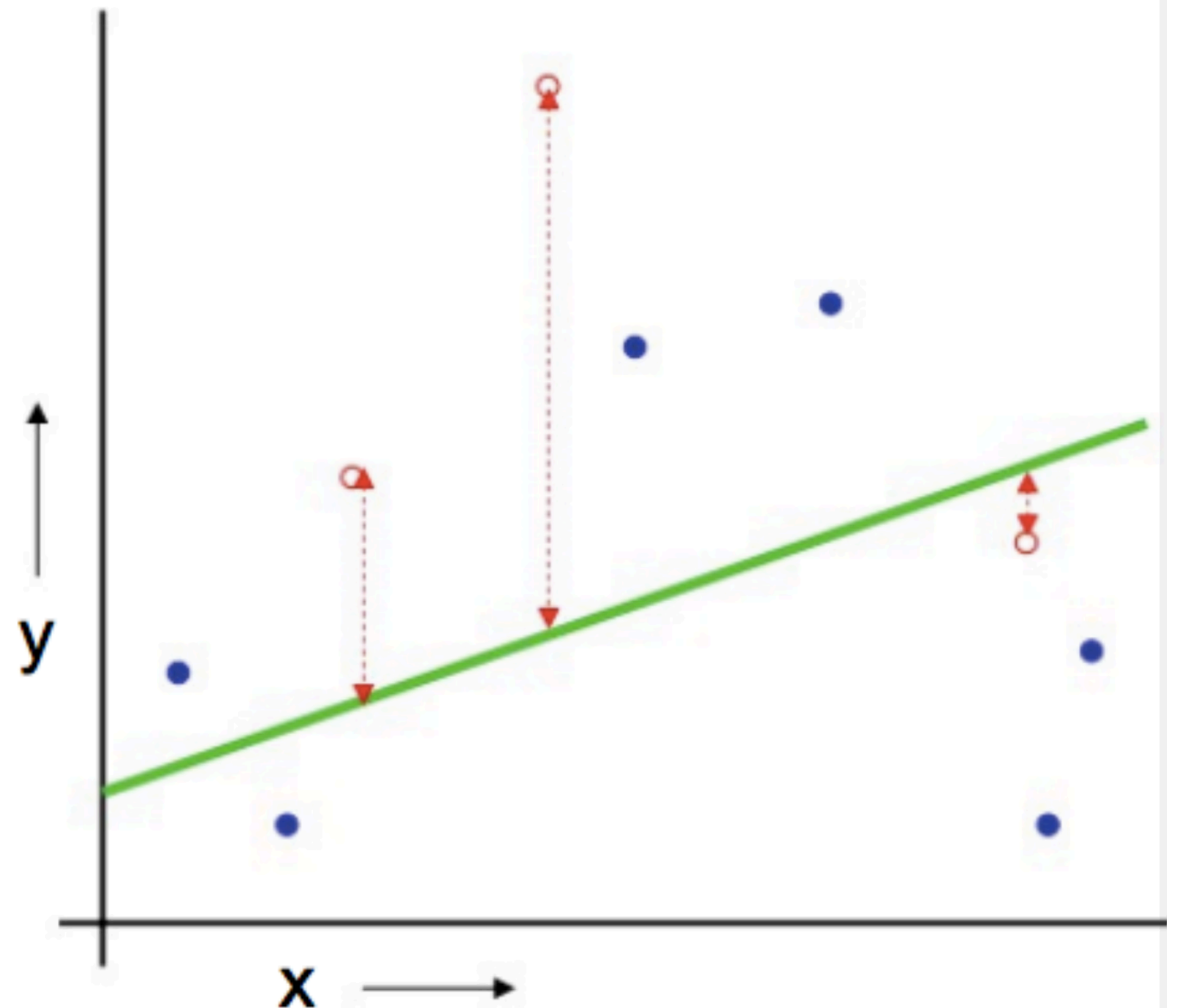


The test set method

Perform your regression on the training set

Estimate your future performance with the test set

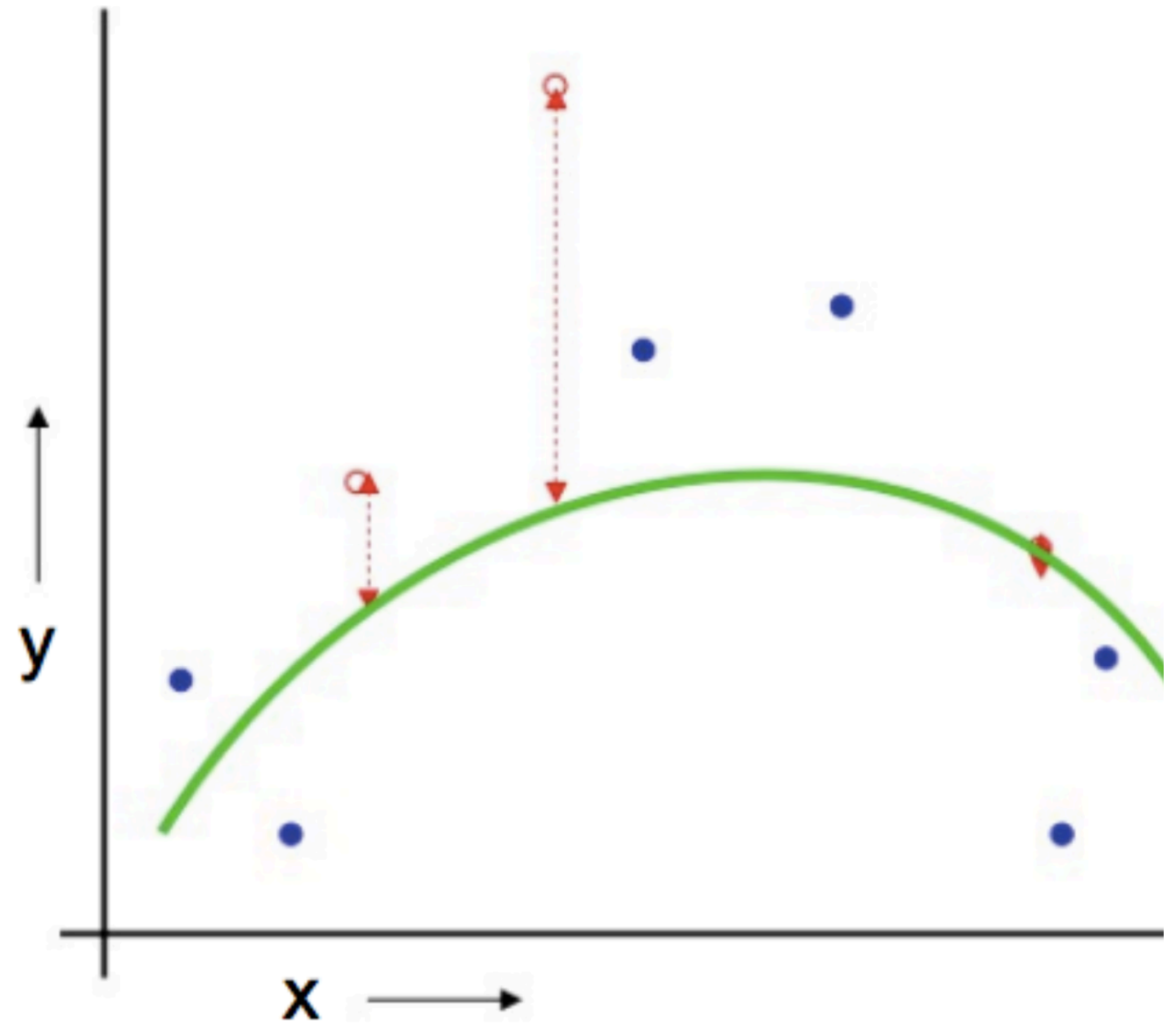
$$MSE = 2.4$$



The test set method

Estimate your future performance with the test set

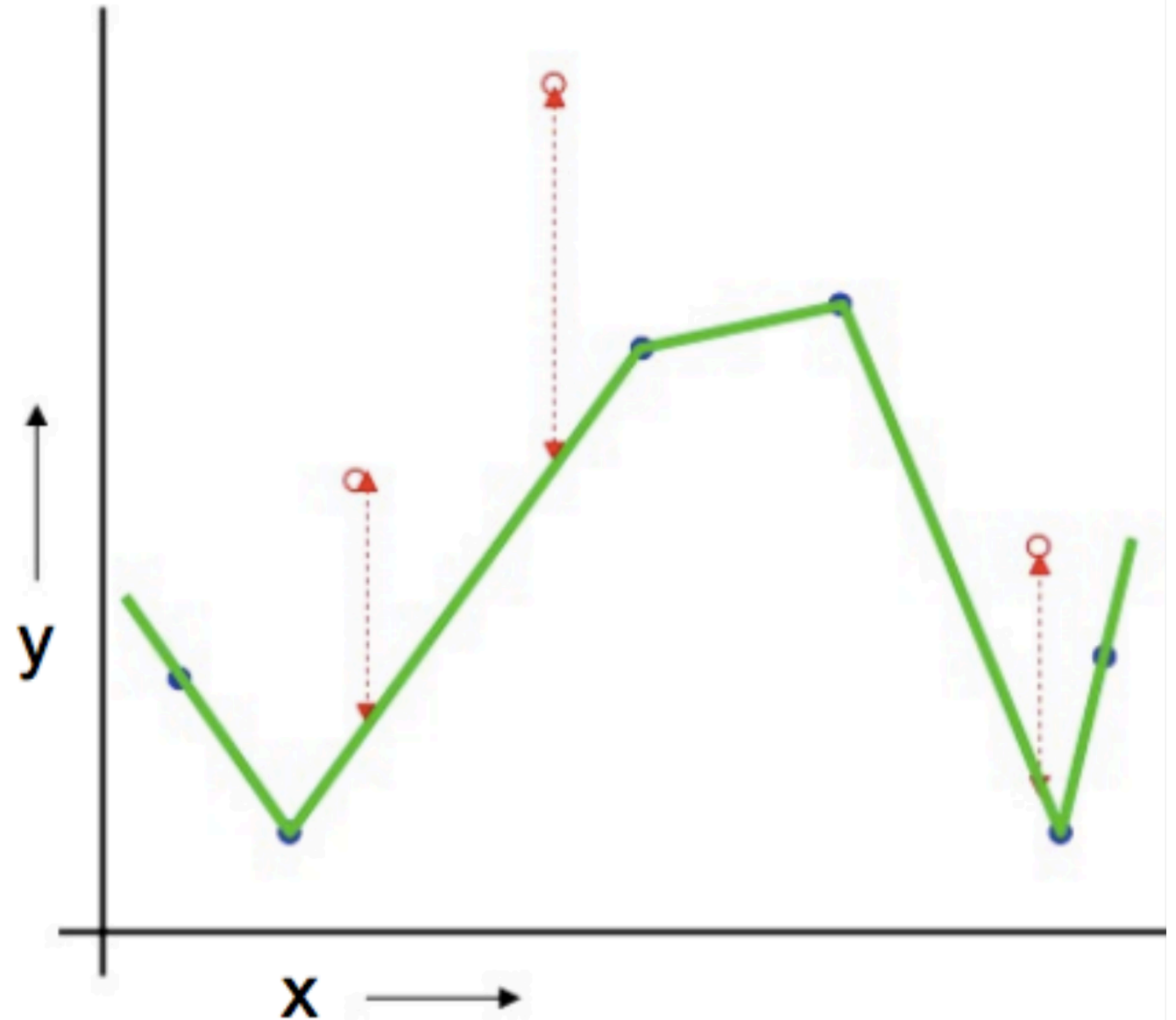
$$MSE = 0.9$$



The test set method

Estimate your future performance with the test set

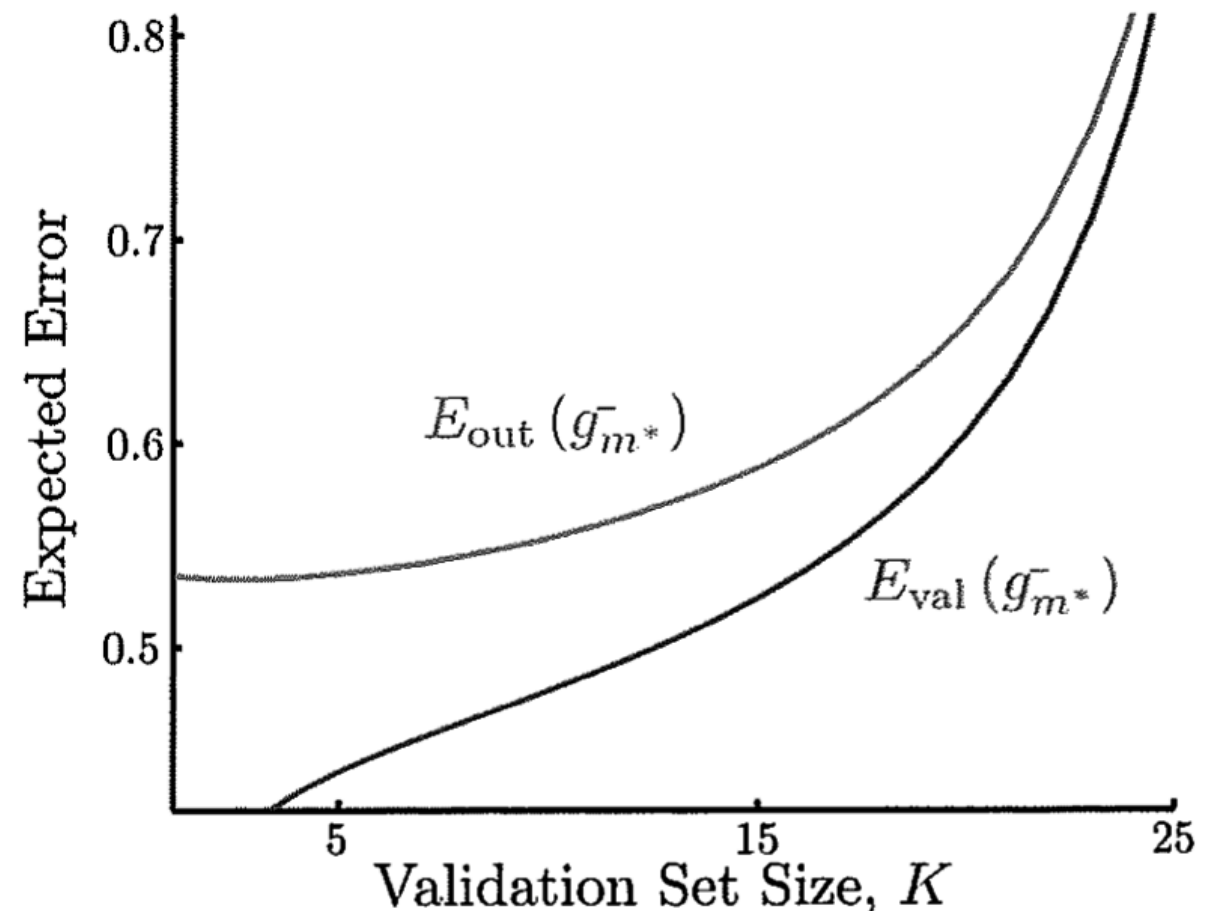
$$MSE = 2.2$$



Model Selection

- Suppose we have M models $\mathcal{H}_1, \dots, \mathcal{H}_M$
- Use validation to select the best model $E_m = E_{\text{val}}(g_m^-); \quad m = 1, \dots, M$
- $\mathcal{H}_{\text{val}} = \{g_1^-, g_2^-, \dots, g_M^-\} \quad |\mathcal{H}_{\text{val}}| = M$

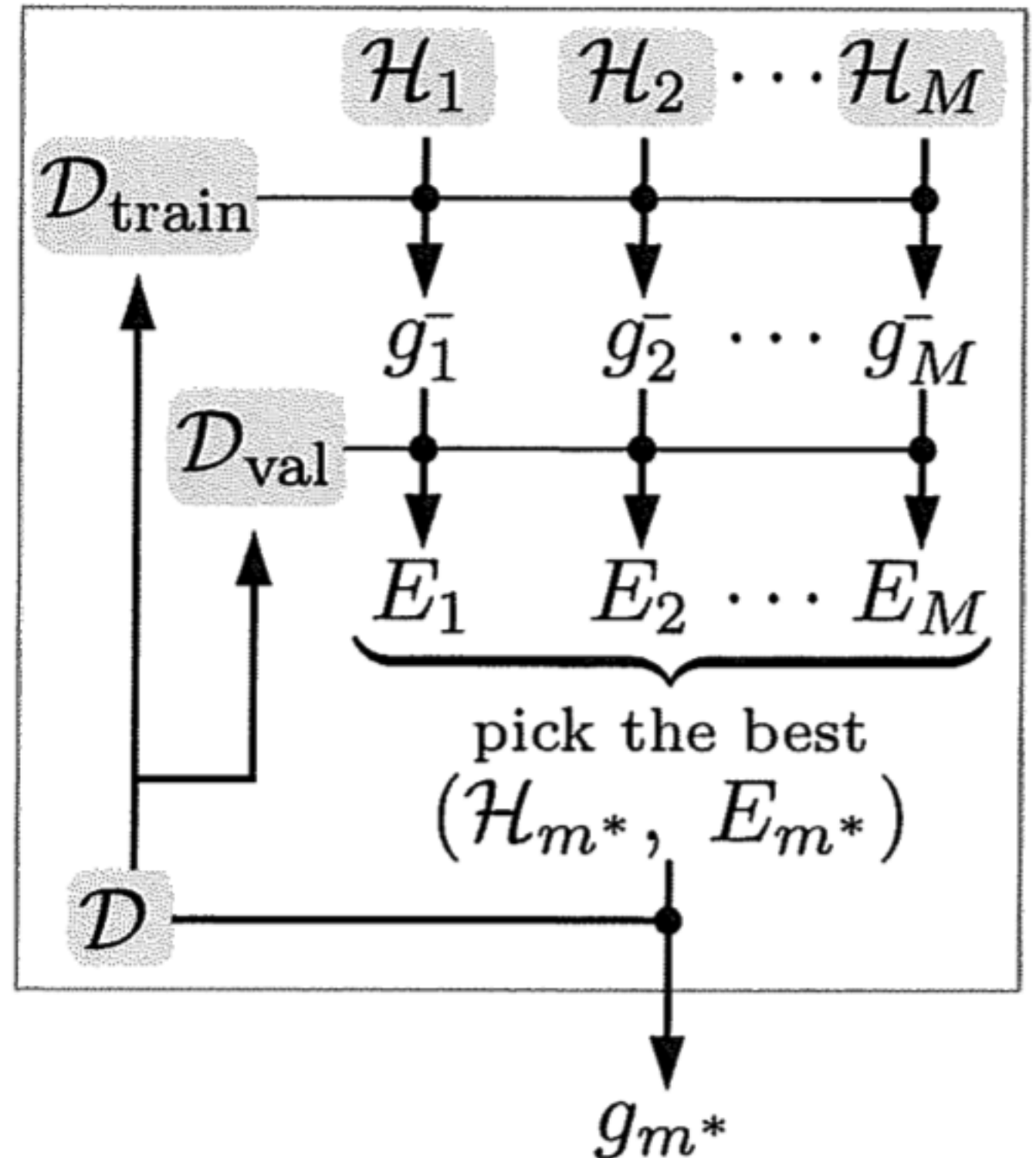
$$E_{\text{out}}(g_{m^*}^-) \leq E_{\text{val}}(g_{m^*}^-) + O\left(\sqrt{\frac{\ln M}{K}}\right)$$



Model Selection

- The goal of model selection is NOT to select g_m^- as our final hypothesis
- The goal is to select the optimal model $\mathcal{H}_m^*$ from

$$\mathcal{H}_1, \dots, \mathcal{H}_M$$



$$E_{\text{out}}(g_{m^*}) \leq E_{\text{out}}(g_{m^*}^-) \leq E_{\text{val}}(g_{m^*}^-) + O\left(\sqrt{\frac{\ln M}{K}}\right)$$

Model Selection

- The goal is to select the optimal model $(\mathcal{H}_m^*, \lambda_m^*)$ from

$$(\mathcal{H}, \lambda_1), (\mathcal{H}, \lambda_2), \dots, (\mathcal{H}, \lambda_M)$$

$$E_{\text{out}}(g_{m^*}) \leq E_{\text{out}}(g_{m^*}^-) \leq E_{\text{val}}(g_{m^*}^-) + O\left(\sqrt{\frac{\ln M}{K}}\right)$$

$$\lambda_1 = 0, \lambda_2 = 0.01, \lambda_3 = 0.02, \dots, \lambda_M = 10$$

- Should we “fine tune” the model? i.e., select model from a very large M ? Does it matter if we choose a large M ?
- Can we do model selection for infinite M ?

Cross Validation

- The idea of validation is built on

$$\begin{array}{ccc} E_{\text{out}}(g) & \approx & E_{\text{out}}(g^-) \approx E_{\text{val}}(g^-) \\ \text{(small } K) & & \text{(large } K) \end{array}$$

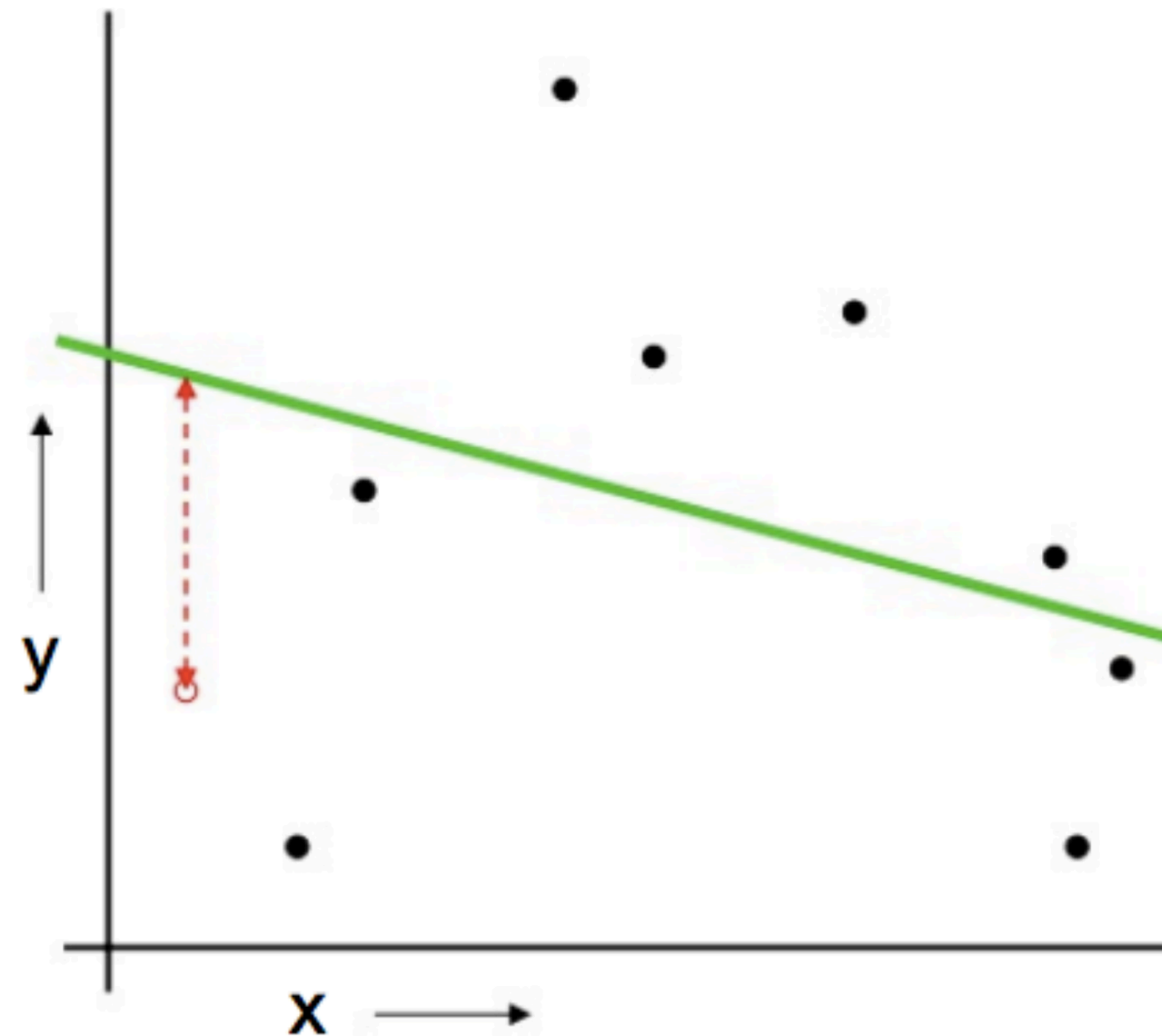
- Key question: should we pick small K or large K ?
- The extreme case: $K = 1$. However, in this case, the RHS estimate is totally useless
- Why? $E_{\text{val}}(g^-)$ will still be an unbiased estimate of $E_{\text{out}}(g^-)$
- So what is the problem? — we can't control the variance! for large probability the RHS is useless (though on average, it is accurate)

LOOCV (留一验证 Leave-one-out cross validation)

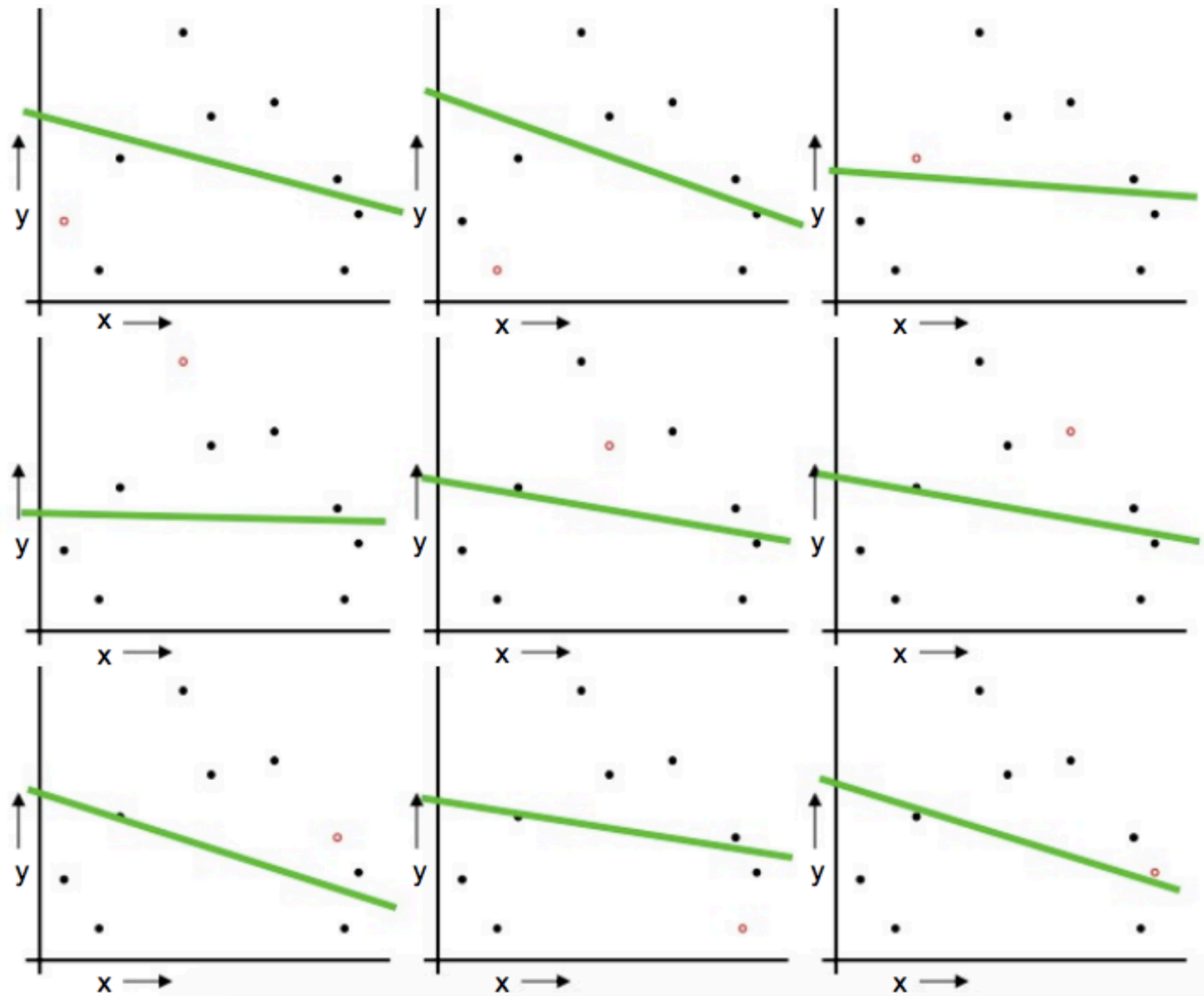
LpOCV, 留 p 验证, $p = 1$

For $k = 1$ to m

- ▶ Choose (x^k, y^k)
- ▶ Temporarily remove (x^k, y^k) from the dataset
- ▶ Training on the remaining $m - 1$ datapoints
- ▶ Compute your error on (x^k, y^k)
- ▶ When you've done all points, report the mean error

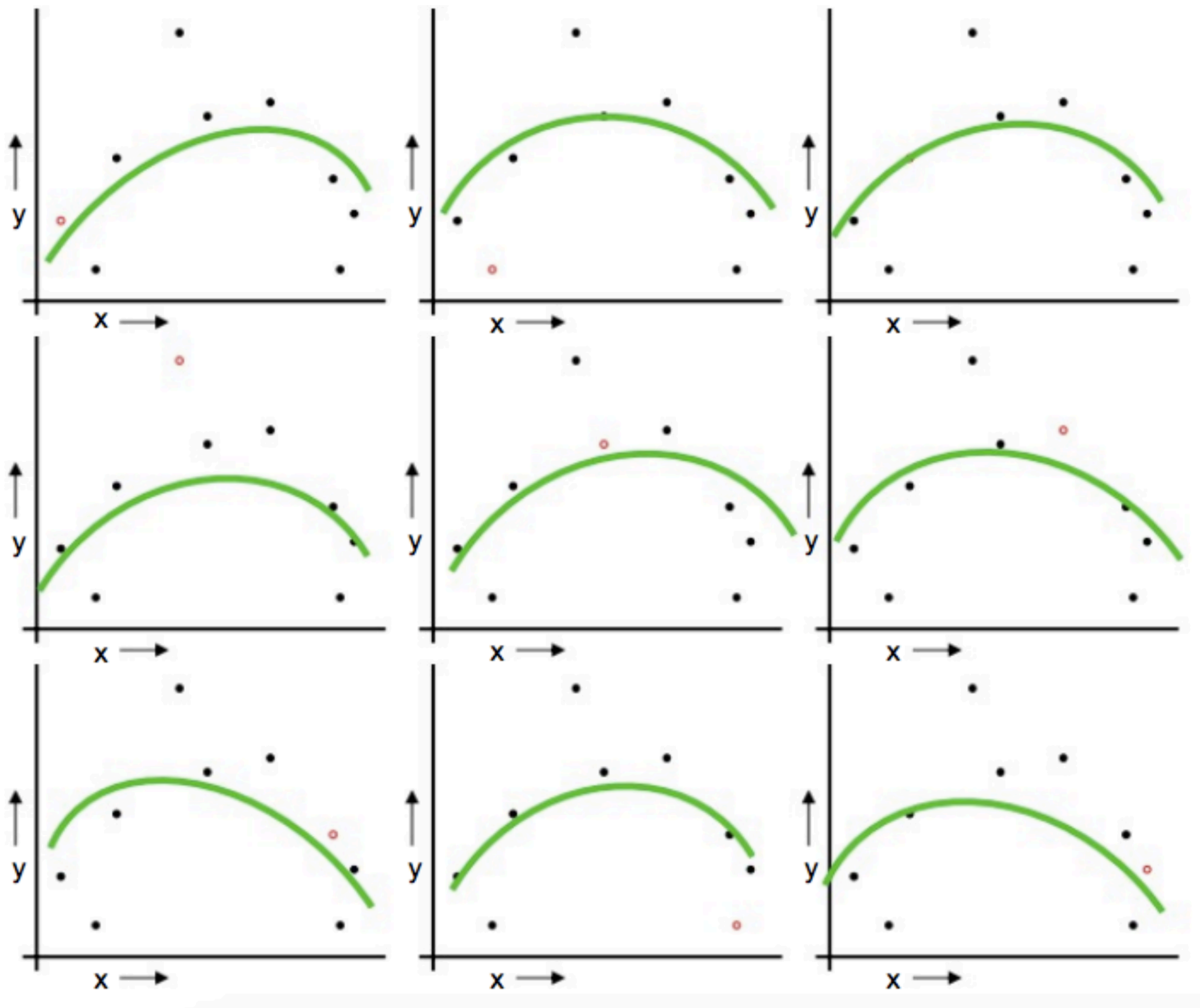


LOOCV (Leave-one-out cross validation)



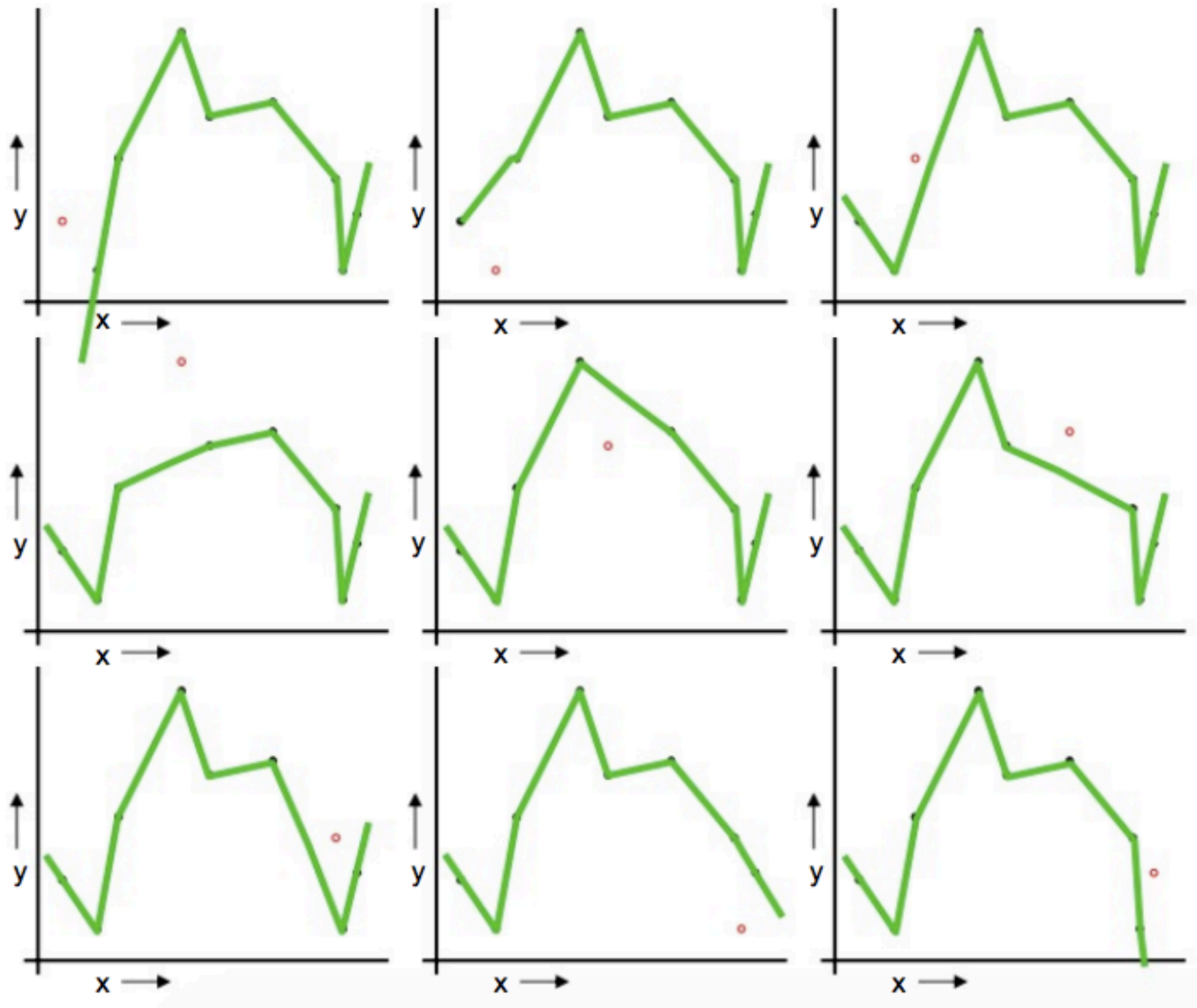
$$\text{MSE}_{loocv} = 2.12$$

LOOCV (Leave-one-out cross validation)



$$\text{MSE}_{loocv} = 0.962$$

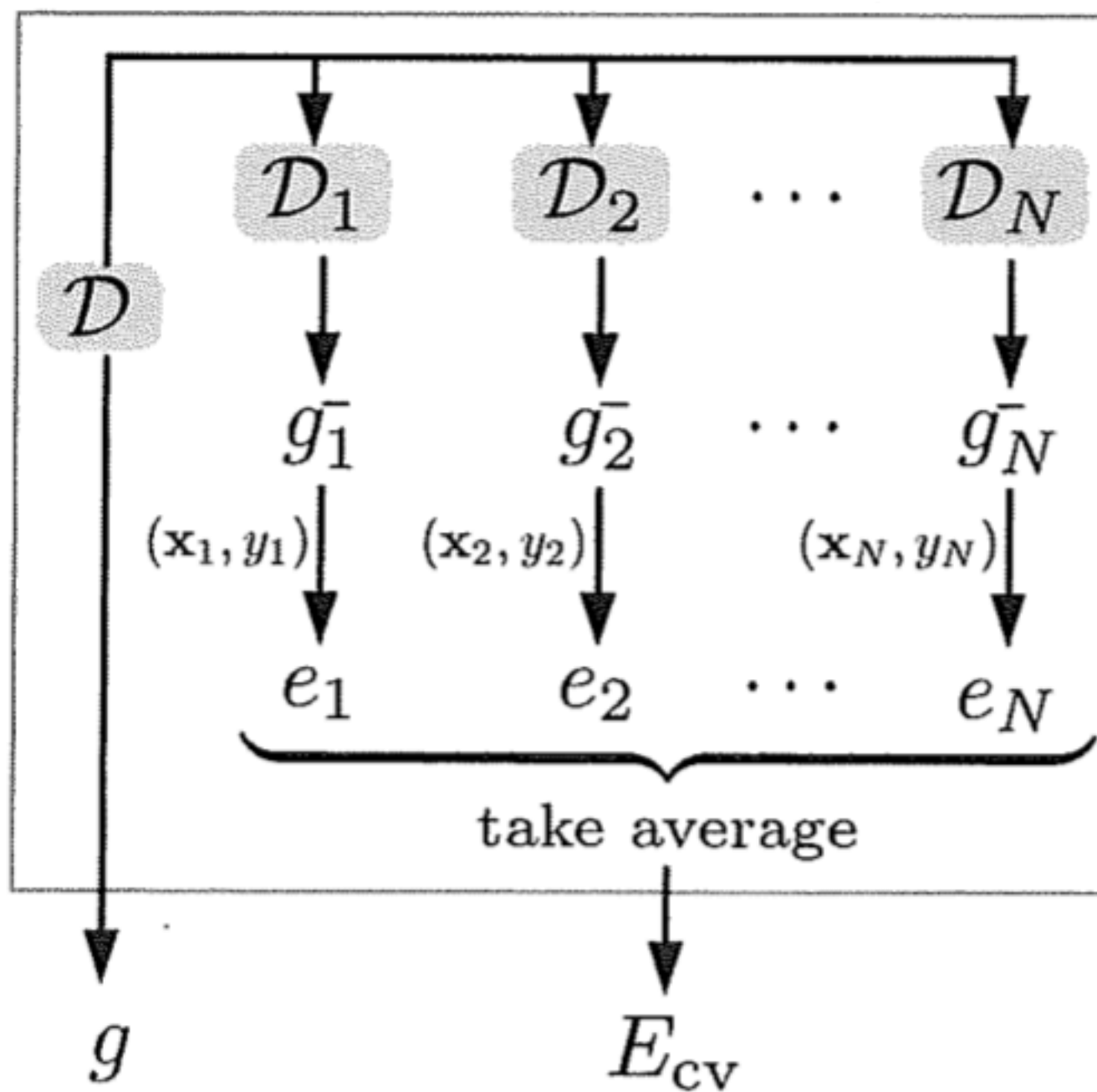
LOOCV (Leave-one-out cross validation)



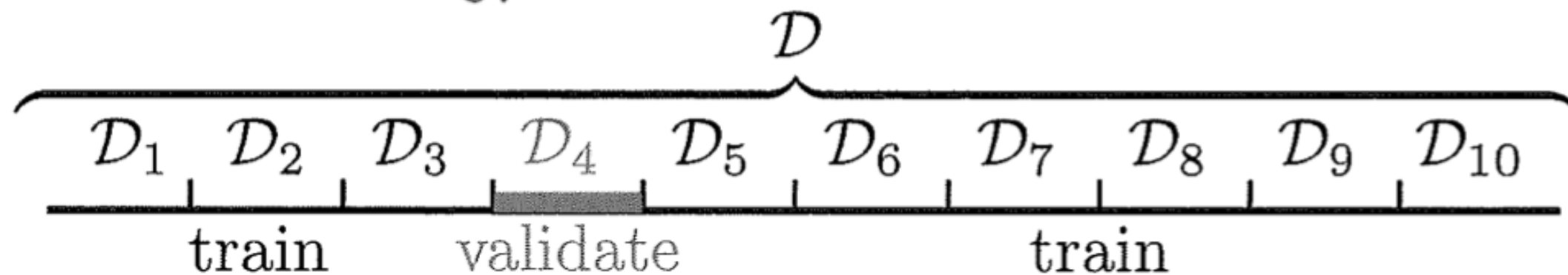
$$\text{MSE}_{loocv} = 3.33$$

Cross Validation

Figure 4.14

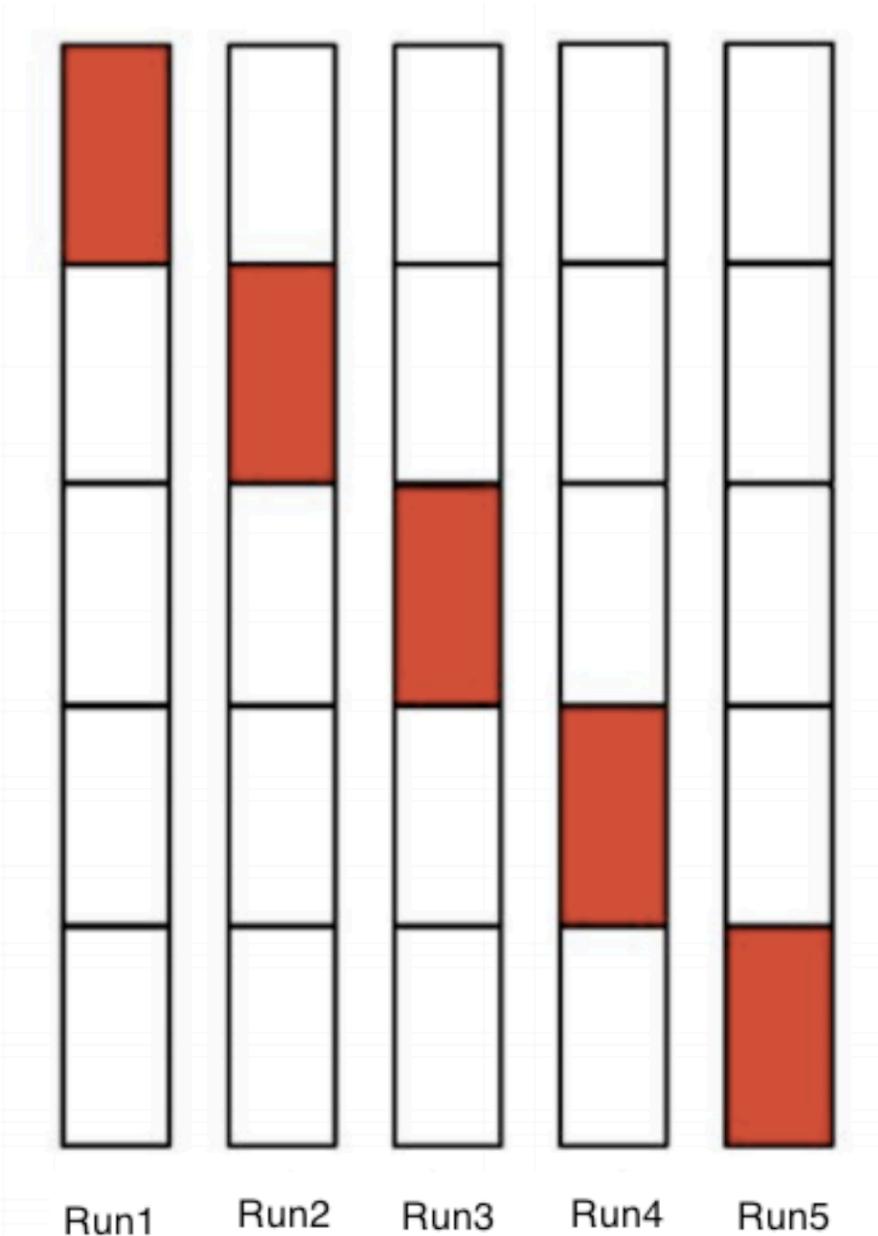


$$E_{\text{out}}(g) \leq E_{\text{cv}}$$



k -fold cross validation

- ▶ Divide data into k blocks
- ▶ For $j = 1$ to k
training on blocks except k -th block, test on k -th block
average the results
- ▶ choose minimum error
- ▶ when $k = m$



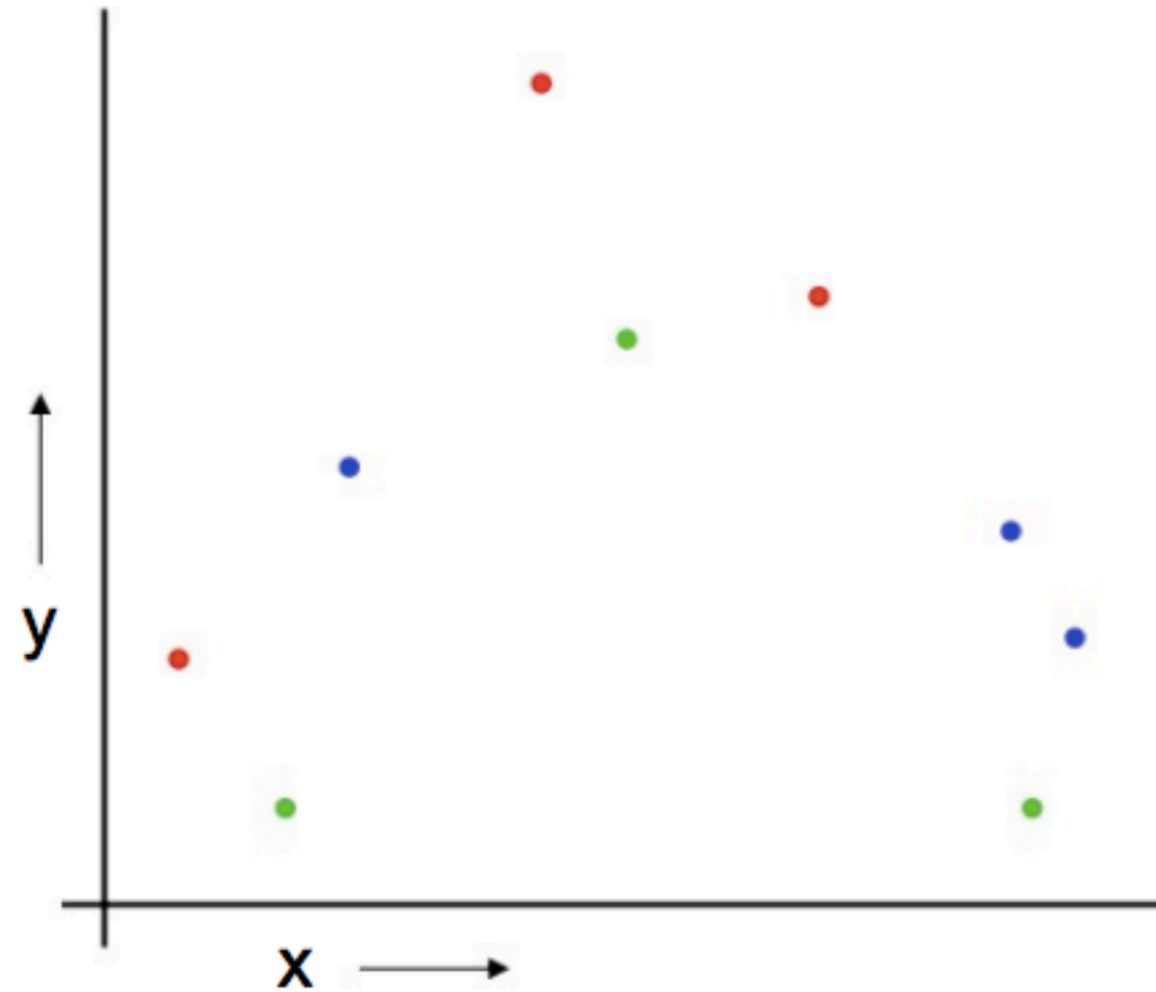
Cross Validation

Cross validation for selecting λ :

- 1: Define M models by choosing different values for λ in the augmented error: $(\mathcal{H}, \lambda_1), (\mathcal{H}, \lambda_2), \dots, (\mathcal{H}, \lambda_M)$
- 2: **for** each model $m = 1, \dots, M$ **do**
- 3: Use the cross validation module in Figure 4.14 to estimate $E_{cv}(m)$, the cross validation error for model m .
- 4: Select the model m^* with minimum $E_{cv}(m^*)$.
- 5: Use model $(\mathcal{H}, \lambda_{m^*})$ and all the data $\mathcal{D}$ to obtain the final hypothesis g_{m^*} . Effectively, you have estimated the optimal λ .

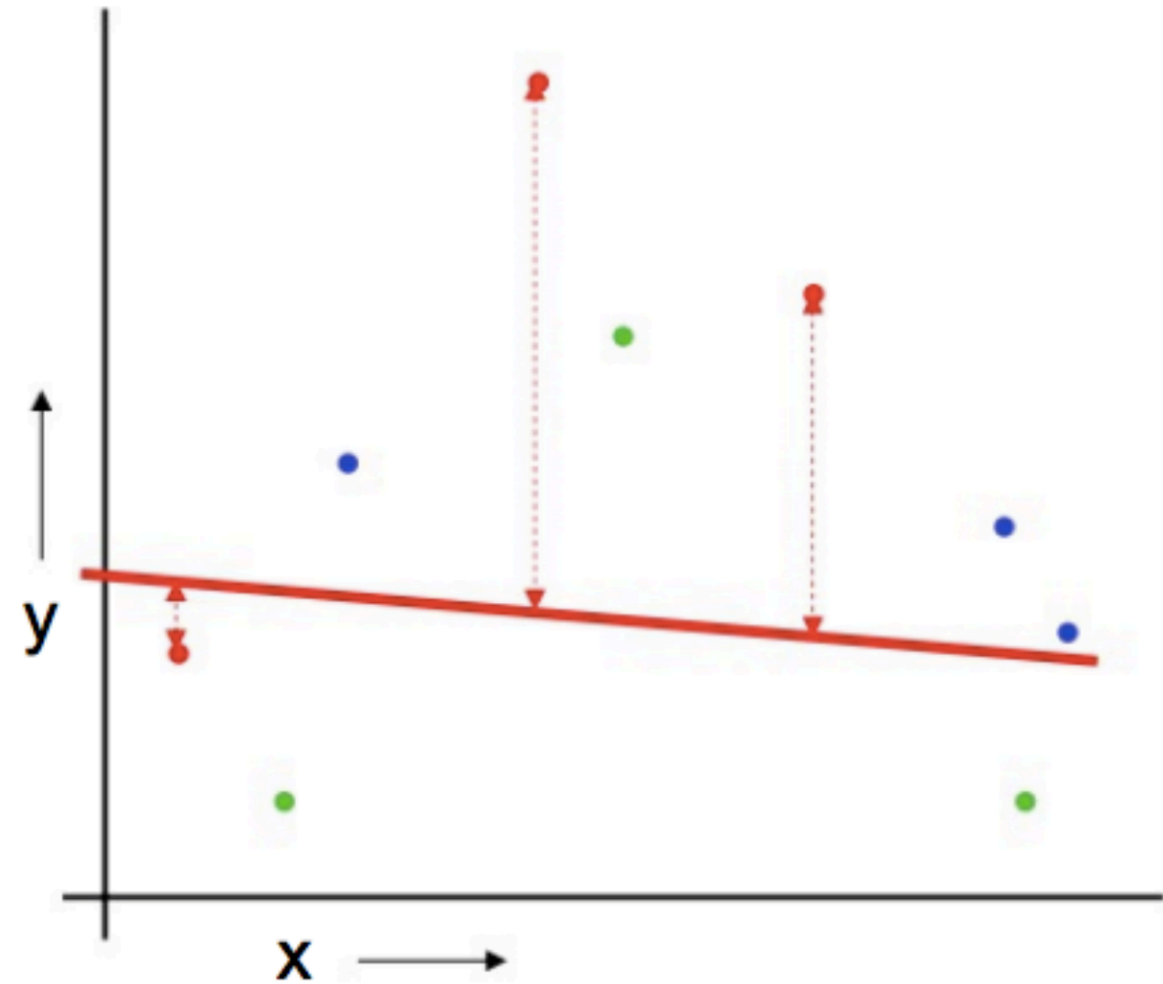
k -fold cross validation

- ▶ In our example, we have $k = 3$
- ▶ Red, Green, and Blue



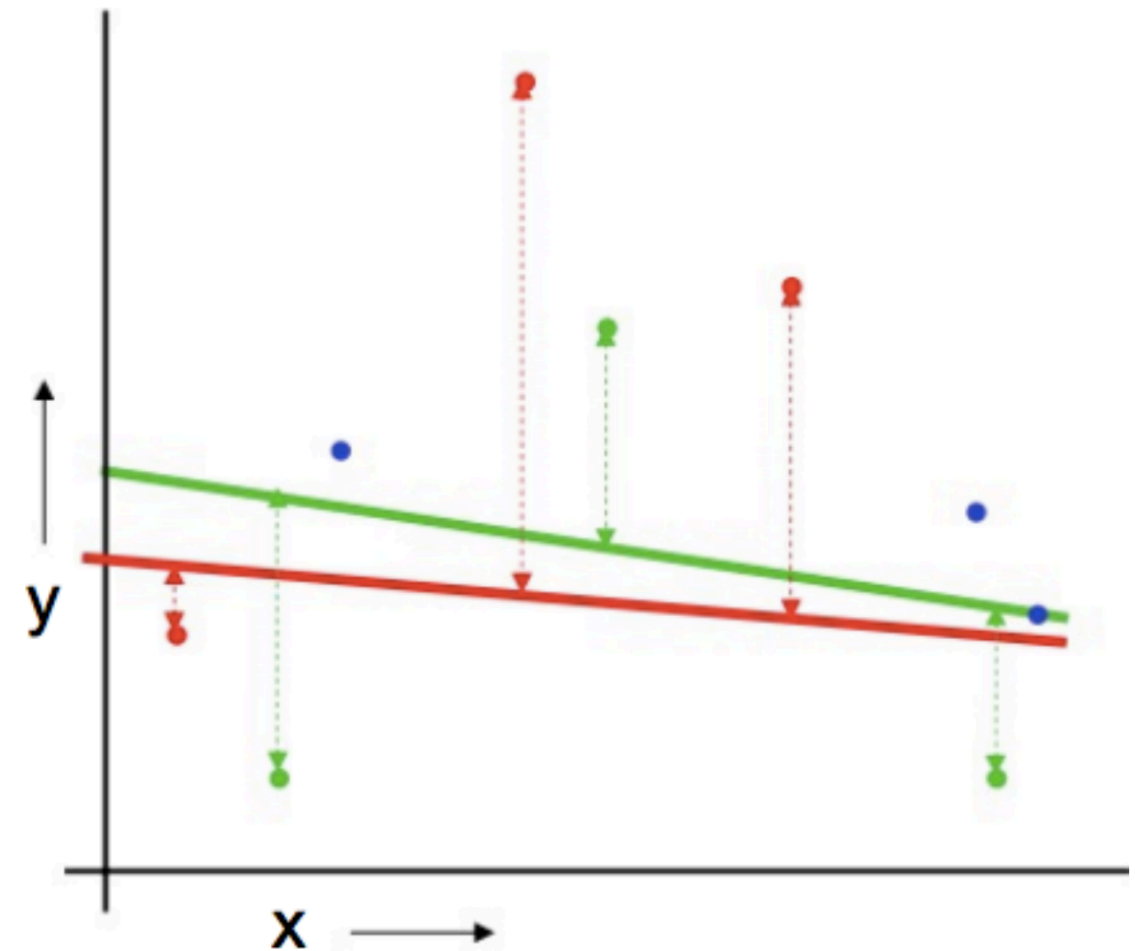
k -fold cross validation

- For the red partition: train on all the points not in the red partition. Find the test-set sum of errors on the red points



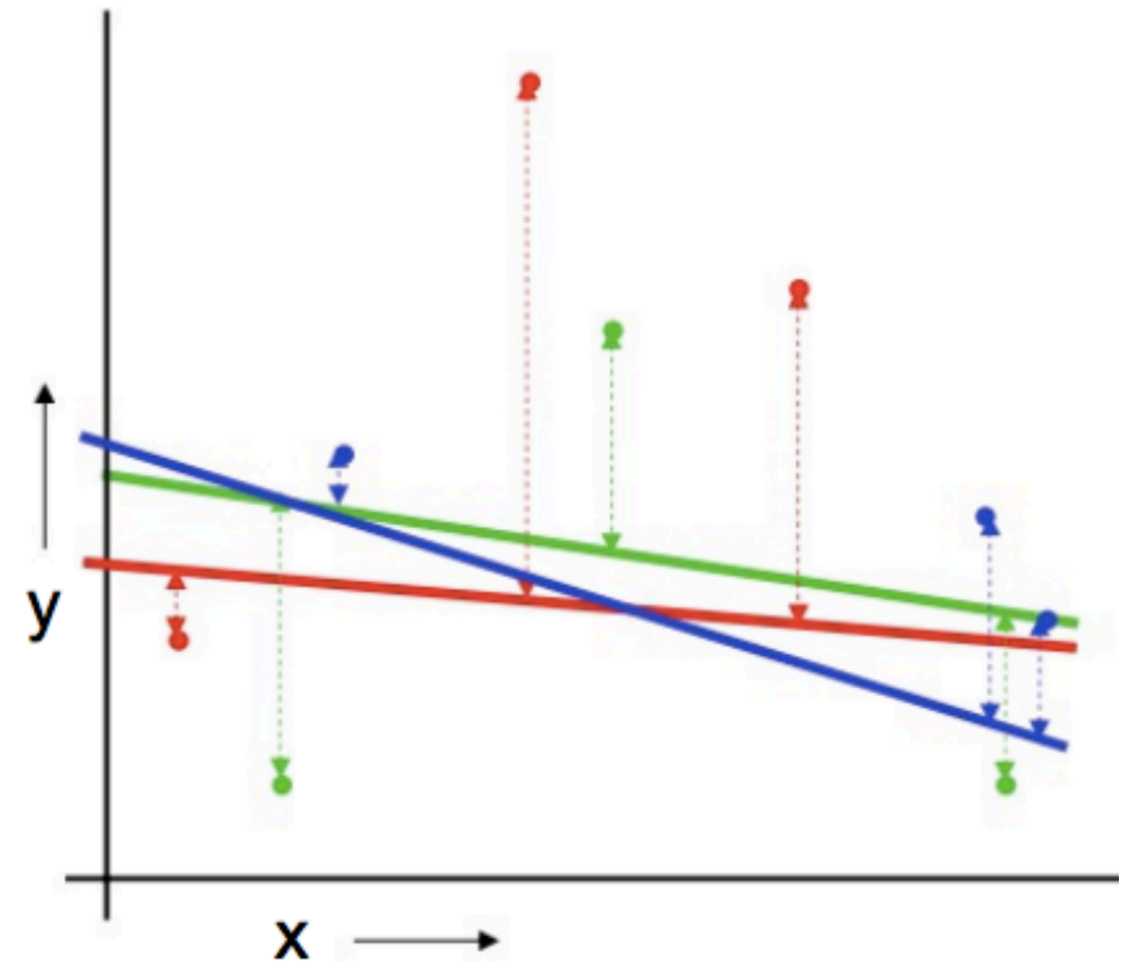
k -fold cross validation

- ▶ For the red partition: train on all the points not in the red partition. Find the test-set sum of errors on the red points
- ▶ For the green partition: train on all the points not in the green partition. Find the test-set sum of errors on the green points



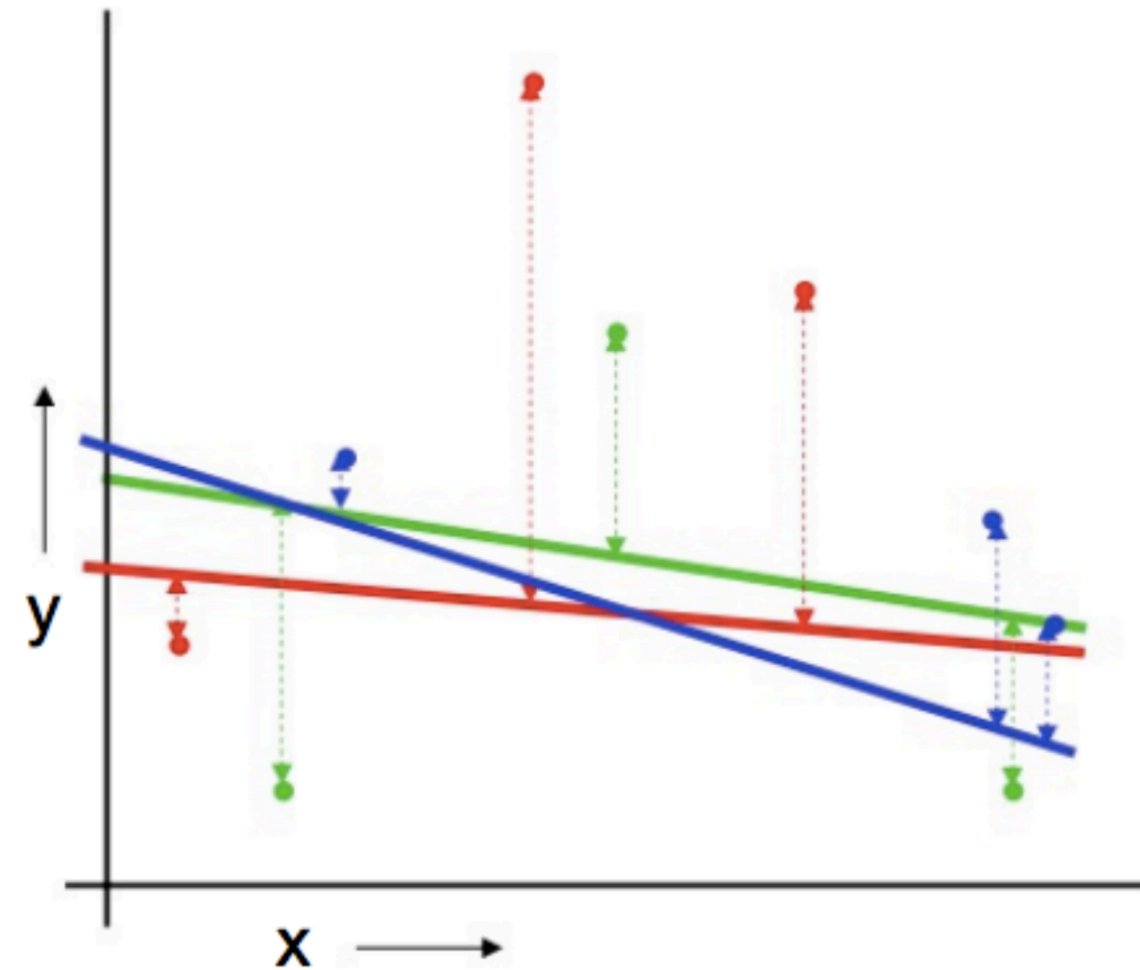
k -fold cross validation

- ▶ For the red partition: train on all the points not in the red partition. Find the test-set sum of errors on the red points
- ▶ For the green partition: train on all the points not in the green partition. Find the test-set sum of errors on the green points
- ▶ For the blue partition: train on all the points not in the blue partition. Find the test-set sum of errors on the blue points



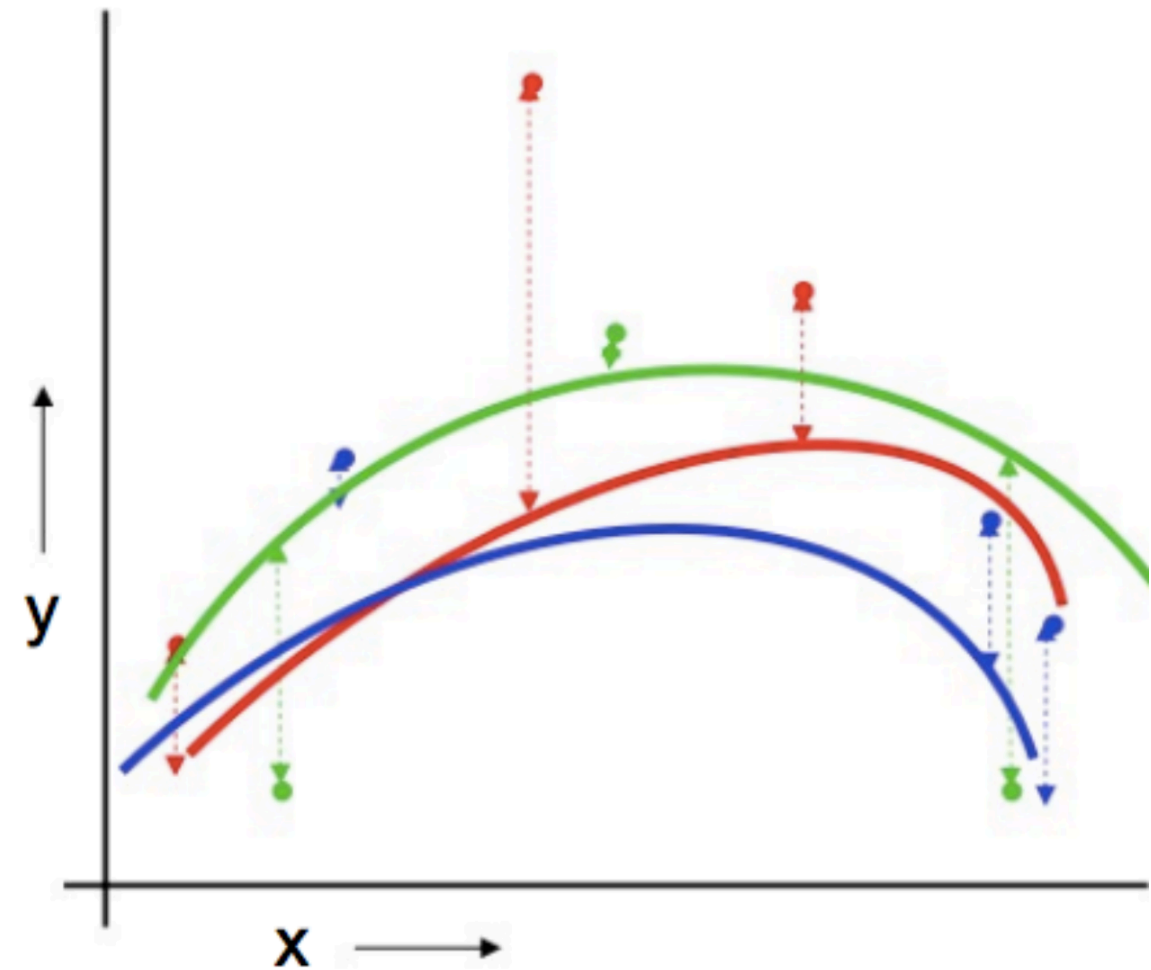
k -fold cross validation

- ▶ Report the mean error
Linear Regression:
 $\text{MSE}_{3\text{-fold}} = 2.05$



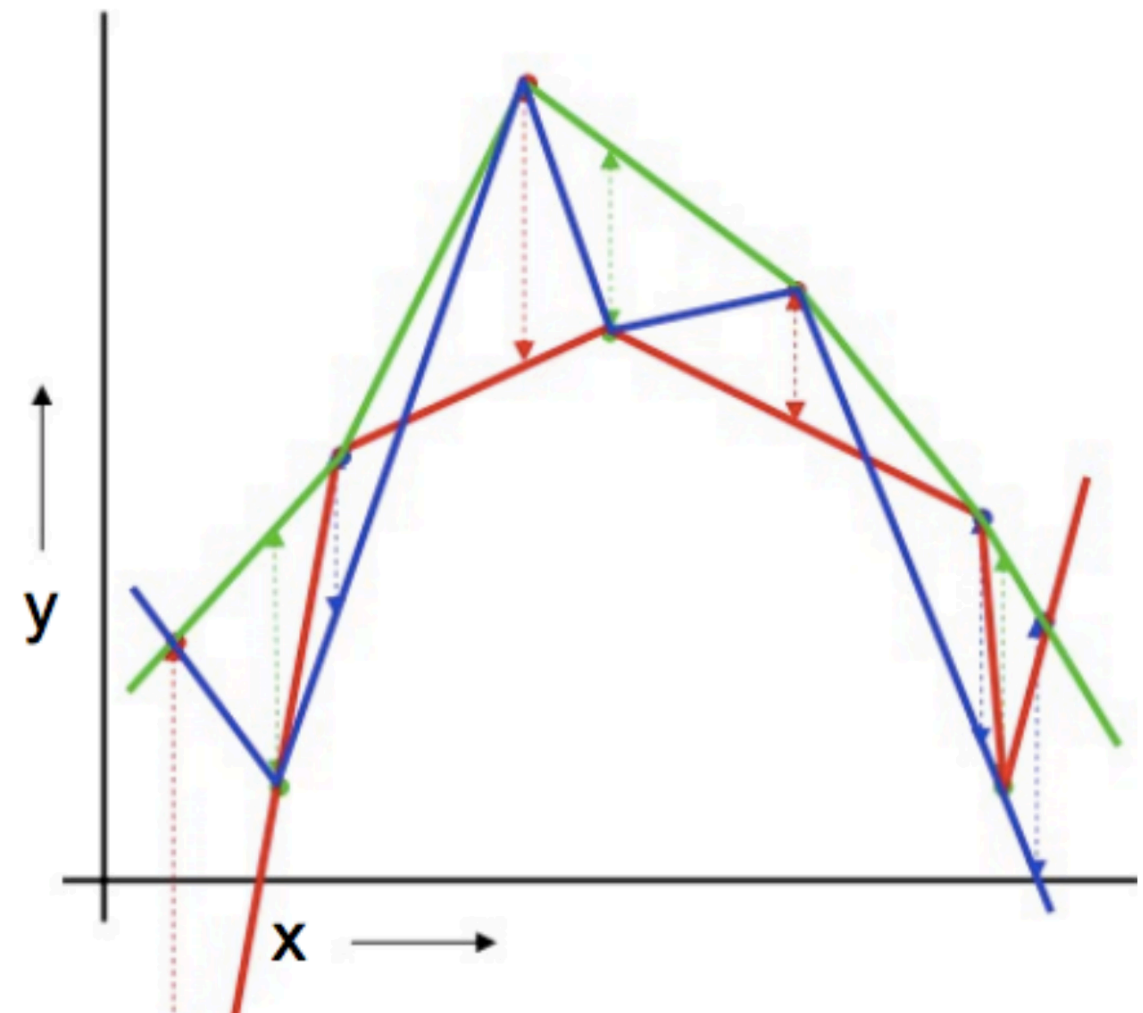
k -fold cross validation

- ▶ Report the mean error
Quadratic Regression:
 $\text{MSE}_{3\text{-fold}} = 1.11$

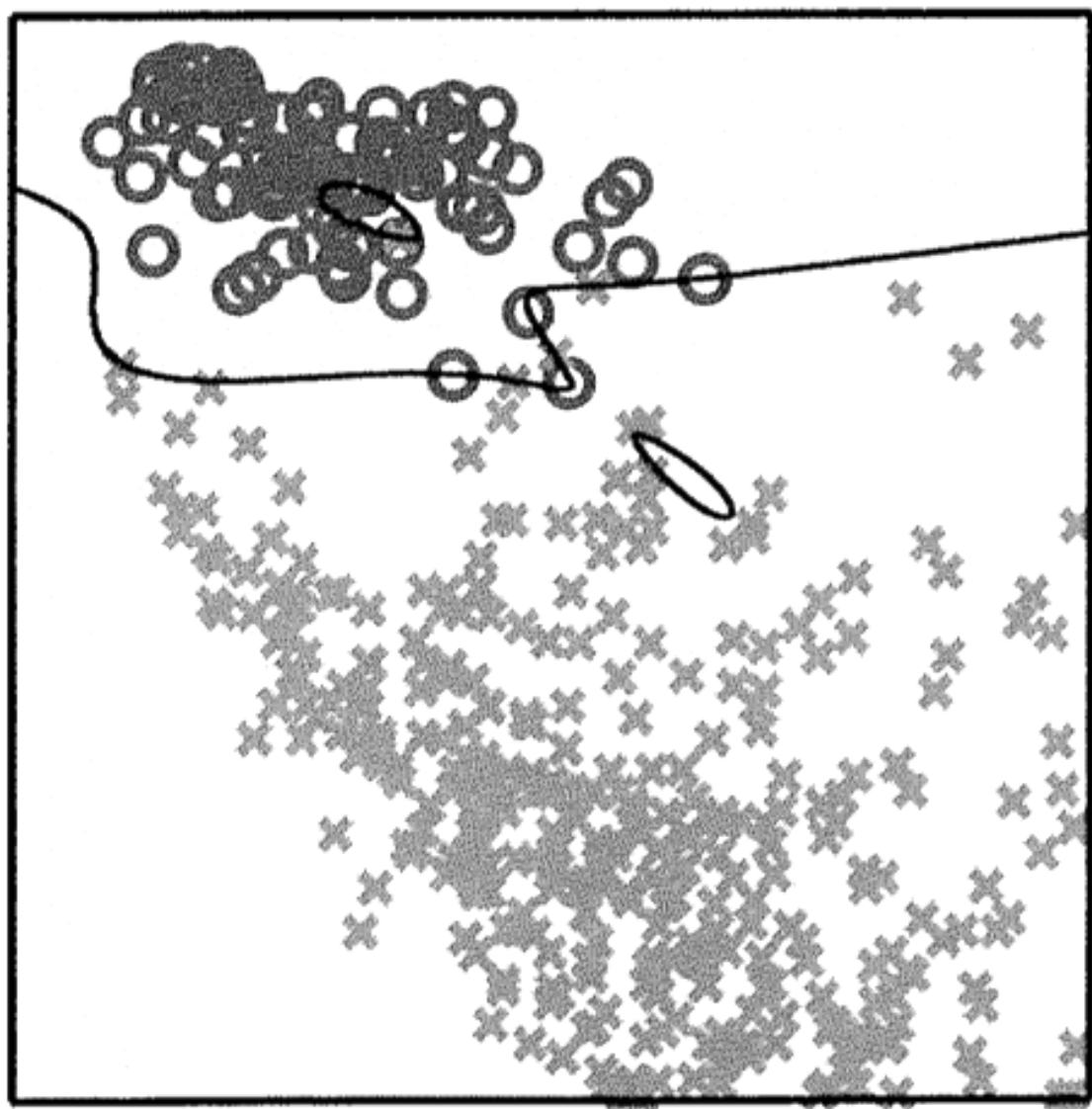


k -fold cross validation

- Report the mean error
Piecewise-linear Regression:
 $\text{MSE}_{3\text{-fold}} = 2.93$



Symmetry



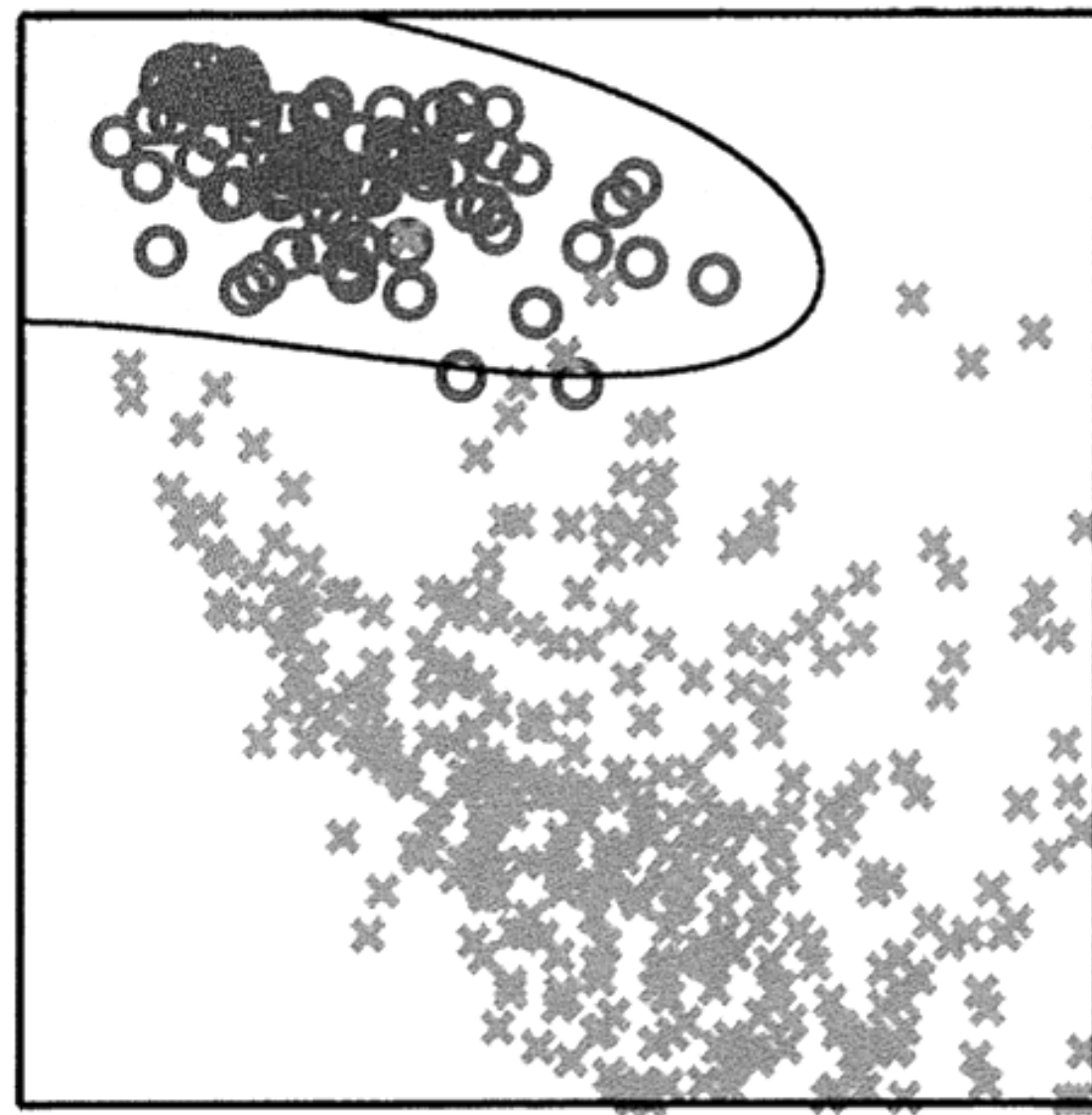
Average Intensity

20-dim classifier (no validation)

$$E_{\text{in}} = 0\%$$

$$E_{\text{out}} = 2.5\%$$

Symmetry



Average Intensity

6-dim classifier (LOO-CV)

$$E_{\text{in}} = 0.8\%$$

$$E_{\text{out}} = 1.5\%$$

1. Noise (stochastic or deterministic) affects learning adversely, leading to overfitting.
2. Regularization helps to prevent overfitting by constraining the model, reducing the impact of the noise, while still giving us flexibility to fit the data.
3. Validation and cross validation are useful techniques for estimating E_{out} . One important use of validation is model selection, in particular to estimate the amount of regularization to use.